

Ex. Nos. 6	Snort IDS/IPS to detect and prevent real time threats in TryHackMe Platform
Date :	

AIM:

To detect and prevent real-time network attacks using Snort IDS/IPS on the TryHackMe platform.

SOFTWARE REQUIRED:

TryHackMe account, AttackBox or VPN, Snort, Nmap, Linux terminal. ***HOW TO***

START TRYHACKME

1. Login to TryHackMe.
2. Search for "Snort" room.
3. Click Start Machine.
4. Click Start AttackBox.
5. Open Terminal in AttackBox.
6. Verify Snort:

```
snort -V
```

7. Test configuration:

```
sudo snort -T -c /etc/snort/snort.conf
```

Experiment 1 — ICMP (Ping) Detection (IDS) — line by line (with new terminal steps)

1. Open terminal and Check Snort:

```
snort -V
```

2. Test Configuration

```
sudo snort -T -c /etc/snort/snort.conf
```

3. Open rule File

```
sudo nano /etc/snort/rules/local.rules
```

4. Add this rule:(ENSURE ONLY THIS RULE EXISTS)

```
alert icmp any any -> any any (msg:"ICMP Ping Detected";
sid:1000001; rev:1;)
```

5. Save and exit: Ctrl+O → Enter → Ctrl+X

6. Run Snort (IDS Mode)

```
sudo snort -c /etc/snort/snort.conf -i eth0
```

7. Generate Attack (New Terminal) using Open new terminal In AttackBox:

- Click + (New Tab) at top of terminal
OR
- Right-click terminal → New Tab

8. In the new terminal:(ATTACK BOX IP ADDRESS) ping
10.48.190.168

9. Go back to first terminal and View Alert → confirm alert shown ICMP Ping Detected

10.(Log check – optional):

```
sudo tail -f /var/log/snort/alert
```

```
ubuntu@ip-10-49-182-45:~$ snort -v
Running in packet dump mode

--== Initializing Snort ==--
Initializing Output Plugins!
pcap DAQ configured to passive.
Acquiring network traffic from "eth0".
ERROR: Can't start DAQ (-1) - socket: Operation not permitted!
Fatal Error, Quitting..
ubuntu@ip-10-49-182-45:~$ sudo snort -T -c /etc/snort/snort.conf
Running in Test mode

--== Initializing Snort ==--
Initializing Output Plugins!
Initializing Preprocessors!
Initializing Plug-ins!
Parsing Rules file "/etc/snort/snort.conf"
PortVar 'HTTP_PORTS' defined : [ 80:81 311 383 591 593 901 1220 1414 1741 1830 23
2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001 7144:7145 7510 7777 7779
000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8
8 8899 9000 9060 9080 9090:9091 9443 9999 11371 34443:34444 41080 50002 55555 ]
PortVar 'SHELLCODE_PORTS' defined : [ 0:79 81:65535 ]
PortVar 'ORACLE_PORTS' defined : [ 1024:65535 ]
```

```

Rules Engine: SF_SNORT DETECTION_ENGINE Version 2.4 <Build 1>
Preprocessor Object: SF_POP Version 1.0 <Build 1>
Preprocessor Object: SF_SSH Version 1.1 <Build 3>
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>
Preprocessor Object: SF_DNS Version 1.1 <Build 4>
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>
Preprocessor Object: SF_SDF Version 1.1 <Build 1>
Preprocessor Object: SF_DCERPC2 Version 1.0 <Build 3>
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>

```

Snort successfully validated the configuration!

Snort exiting

ubuntu@ip-10-49-182-45:~\$

```

64 bytes from 10.48.190.168: icmp_seq=412 ttl=64 time=0.039 ms
^C
--- 10.48.190.168 ping statistics ---
412 packets transmitted, 412 received, 0% packet loss, time 420857ms
rtt min/avg/max/mdev = 0.005/0.037/0.048/0.004 ms
ubuntu@ip-10-48-190-168:~$

```

```

Commencing packet processing (pid=1941)
55: Session exceeded configured max segs to queue 2621 using 2621 segs (server queue). 10.48.127.64 4226
4 --> 10.48.190.168 80 (8) : LMstate 0x48 LMflags 0x2101
03/10-10:22:44.966713  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {IPV6-ICMP} fe80::79:32f
f:fe6b:91a7 -> ff02::2

```

Experiment 2 — Port Scan Detection — line by line (with new terminal steps)

1. Edit Rules File

```
sudo nano /etc/snort/rules/local.rules
```

2. Add:

```

alert tcp any any -> any any (msg:"TCP Port Scan Activity";
sid:1000002; rev:1;)

```

3. Save: Ctrl+O → Enter → Ctrl+X

4. Stop Snort:

```
Ctrl + C
```

5. Restart Snort:

```
sudo snort -A console -c /etc/snort/snort.conf -i eth0
```

6. Open new terminal tab

- Click + on terminal
OR
- Right-click → New Tab

7. In new terminal:

for p in {1..100}; do nc -zv 10.48.190.168 \$p; done

8. Go back to Snort terminal → confirm alert

9. (Optional log):

sudo tail -f /var/log/snort/alert

```
03/10-11:03:14.096390 [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] (TCP) 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.096539 [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] (TCP) 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.096994 [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] (TCP) 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.097359 [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] (TCP) 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.097699 [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] (TCP) 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.101333 [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] (TCP) 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.118448 [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] (TCP) 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.118519 [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] (TCP) 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.118767 [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] (TCP) 10.48.127.64:4
2264 -> 10.48.190.168:80
```

```
nc: connect to 10.48.190.168 port 14 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 15 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 16 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 17 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 18 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 19 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 20 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 21 (tcp) failed: Connection refused
Connection to 10.48.190.168 22 port [tcp/ssh] succeeded!
nc: connect to 10.48.190.168 port 23 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 24 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 25 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 26 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 27 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 28 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 29 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 30 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 31 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 32 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 33 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 34 (tcp) failed: Connection refused
```

Experiment 3 — IPS Blocking (Inline Mode) — line by line (with new terminal steps)

1. Change rule in Rule File

sudo nano /etc/snort/rules/local.rules

2. Add:

drop icmp any any -> any any (msg:"ICMP Blocked"; sid:1000003; rev:1;)

3. Save: Ctrl+O → Enter → Ctrl+X

4. Stop Snort:

Ctrl + C

5. Run inline:

sudo snort -Q --daq afpacket -i eth0:eth0 -c /etc/snort/snort.conf 6. Open

new terminal tab

- Click +

OR

- Right-click → New Tab

7. In new terminal:

ping 8.8.8.8

8. Expected result:

No ping reply

Alert shown in Snort terminal

9. (Optional log):

sudo tail -f /var/log/snort/alert

If interface is not eth0

Run once:

ip a

Use the shown interface name instead of eth0.

RESULT:

Successfully detected and prevented attacks using Snort IDS/IPS.